CSC432 – Information Security

Dr. Muhammad Sharjeel

muhammadsharjeel@cuilahore.edu.pk

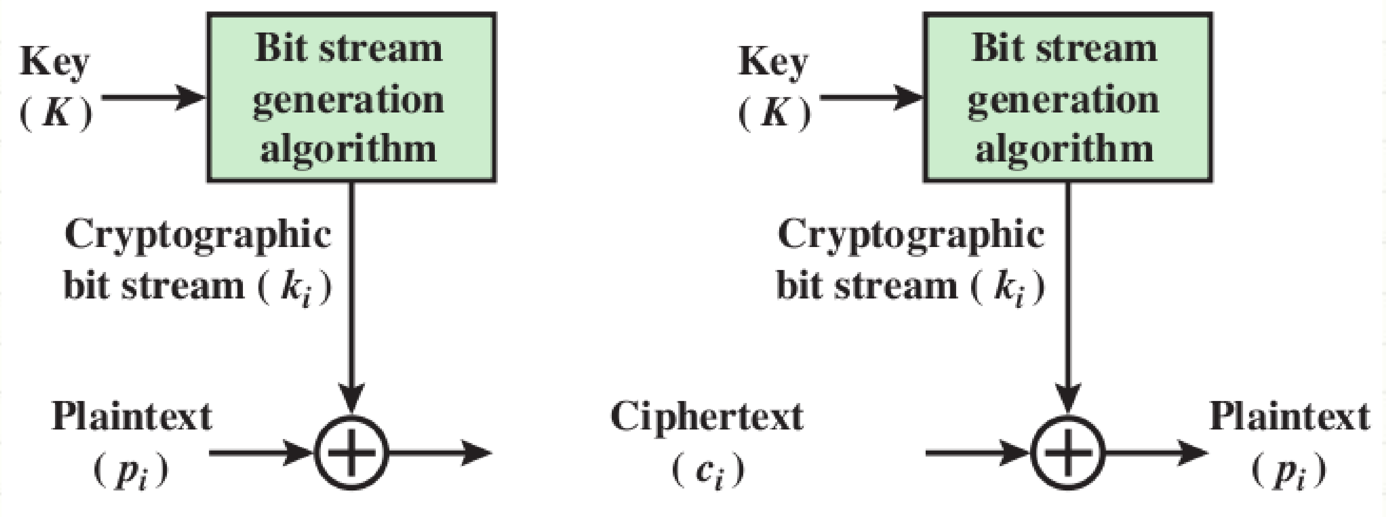
Lecture No 4

BLOCK CIPHERS AND DES

Modern Ciphers

Stream Ciphers

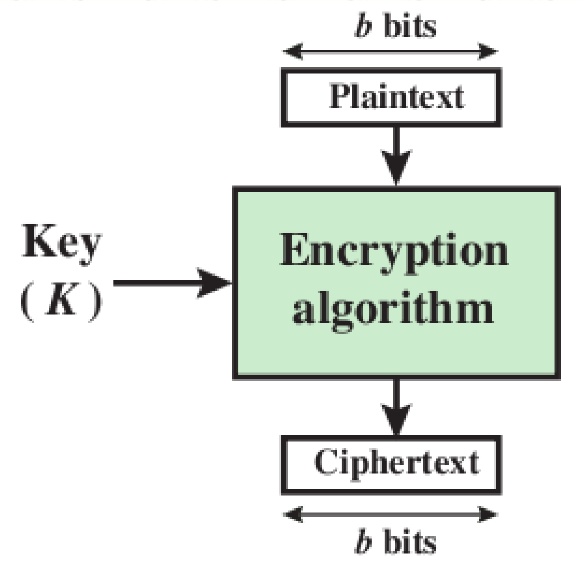
- Encrypts a digital data stream one bit or one byte at a time
- One time pad is an example; but practical limitations
- Typical approach for stream cipher:
  - Key (k) used as input to bit-stream generator algorithm
  - Algorithm generates cryptographic bit stream (k_i) used to encrypt plaintext
  - Users share a key; use it to generate key-stream



Modern Ciphers

Block Ciphers

- Encrypt a block of plaintext as a whole to produce same sized ciphertext
- Typical block sizes are 64 or 128 bits
- Modes of operation used to apply block ciphers to larger plaintexts



Modern Ciphers

Block Ciphers

- ❑ n-bit block cipher takes n-bit plaintext and produces n-bit ciphertext
- ❑ Encryption must be reversible (decryption possible)
- ❑ Each plaintext block must produce unique ciphertext block
- ❑ Total transformations is $2^n!$

Modern Ciphers

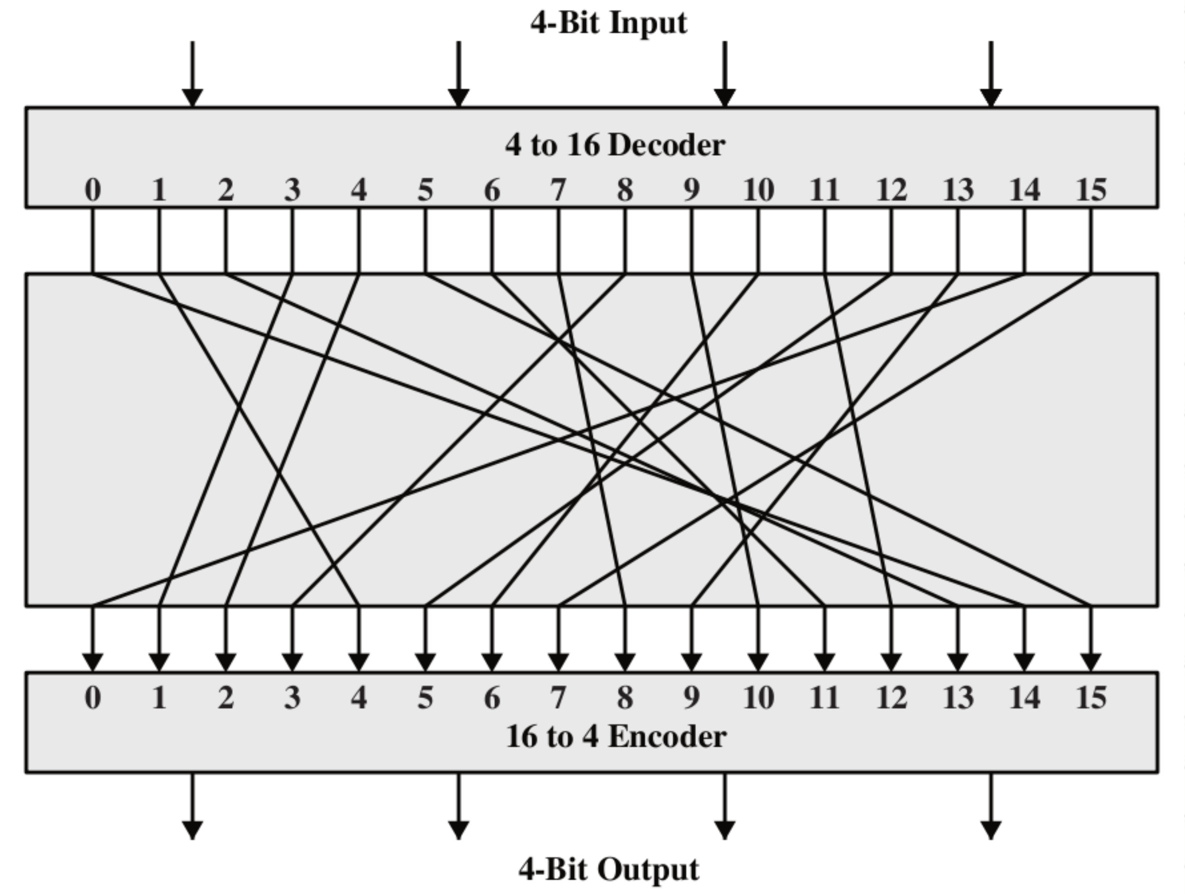
An Ideal Block Cipher

- n-bit input maps to 2^n possible input states
- Substitution used to produce 2^n output states
- Output states map to n-bit output
- An ideal block cipher allows maximum number of possible encryption mappings from plaintext block

Modern Ciphers

Lets see an example

- ❑ 4-bit input = 16 different possible 4-bit patterns



Modern Ciphers

- Encryption key for this ideal block cipher is the codebook itself, meaning the table that shows the relationship between the input blocks and the output blocks

Plaintext	Ciphertext
0000	1110
0001	0100
0010	1101
0011	0001
0100	0010
0101	1111
0110	1011
0111	1000
1000	0011
1001	1010
1010	0110
1011	1100
1100	0101
1101	1001
1110	0000
1111	0111

Ciphertext	Plaintext
0000	1110
0001	0011
0010	0100
0011	1000
0100	0001
0101	1100
0110	1010
0111	1111
1000	0111
1001	1101
1010	1001
1011	0110
1100	1011
1101	0010
1110	0000
1111	0101

Modern Ciphers

<u>An Ideal Block Cipher</u>

- Among all possible mappings, our 'key' here determines the specific mapping
- Using this straightforward method, the key length is 4x16 (4×2^4) = 64

- That implies that the encryption key for the ideal block cipher using 64-bit blocks will be of size $64 \times 2^{64} \approx 10^{21}$
- Don't you think the size of the encryption key would make the ideal block cipher an impractical idea???
  - Think of the logistical issues related to the transmission, storage, and processing of such large keys

Modern Ciphers

An Ideal Block Cipher

- Problems with this ideal block cipher:
  - Small block size:
    - equivalent to classical substitution cipher; cryptanalysis based on statistical characteristics feasible
  - Large block size:
    - key must be very large; performance/implementation problems

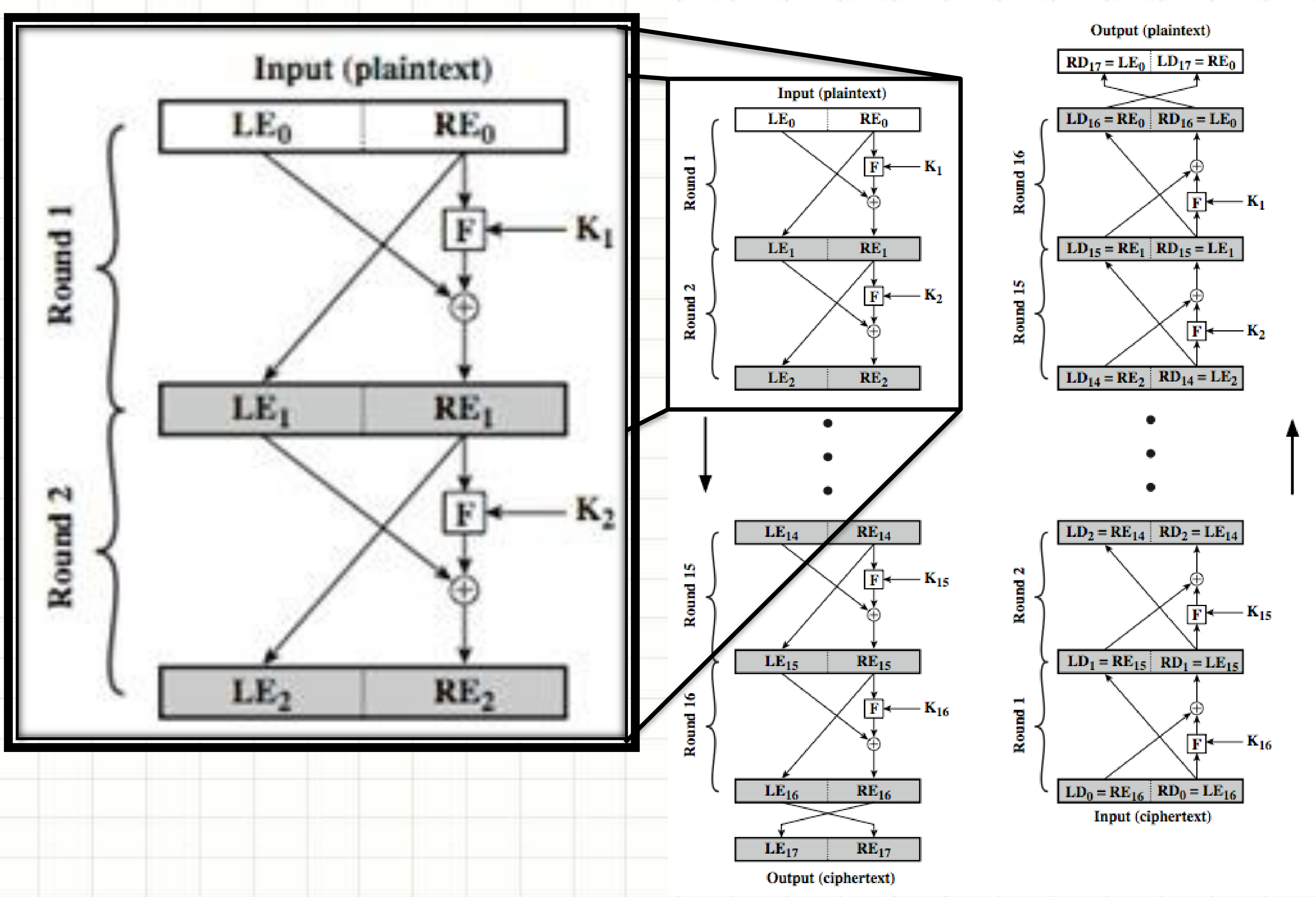
The Feistel Structure for Block Ciphers

- Named after the IBM cryptographer Horst Feistel (early 70's) and first implemented in the Lucifer cipher by Horst Feistel and Don Coppersmith
- Feistel proposed applying two or more simple ciphers in sequence so final result is cryptographically stronger than component ciphers
- n-bit block length; k-bit key length; 2^k transformations
- Feistel cipher alternates: substitutions, transpositions
- Applies concepts of <u>confusion</u> and <u>diffusion</u>
- Applied in many ciphers today

The Feistel Structure for Block Ciphers

- Approach:
  - Plaintext split into two halves
  - Subkeys (or round keys) generated from key
  - Round function, F, applied to right half
  - Apply substitution on left half using XOR
  - Apply permutation by interchanging two halves

The Feistel Structure for Block Ciphers



The Feistel Structure for Block Ciphers

Diffusion

- Statistical nature of plaintext is reduced in ciphertext
- E.g. A plaintext letter affects the value of many ciphertext letters
- How: repeatedly apply permutation (transposition) to data, and then apply function

Confusion

- Make relationship between ciphertext and key as complex as possible
- Even if attacker can find some statistical characteristics of ciphertext, still hard to find key
- How: apply complex (non-linear) substitution algorithm

The Feistel Structure for Block Ciphers

Using the Feistel Structure

- Exact implementation depends on various design features
- Block size, e.g. 64, 128 bits, larger values leads to more diffusion
- Key size, e.g. 128 bits, larger values leads to more confusion, resistance against brute force
- Number of rounds, e.g. 16 rounds
- Subkey generation algorithm, should be complex
- Round function F, should be complex
- Other factors include fast encryption in software and ease of analysis
- Tradeoff: <u>security vs performance</u>

Data Encryption Standard

Data Encryption Standard

- One of most used encryption systems in world
- Designed by IBM (Lucifer) with input from NSA
- Adopted/published by National Bureau of Standards (NBS) now National Institute of Standards and Technology (NIST) in 1977
- For commercial and unclassified government applications
- Efficient hardware implementation
- Electronic payment industry (ATM machines) uses DES
- Microsoft OneNote, Outlook use DES to password protect user content and system data

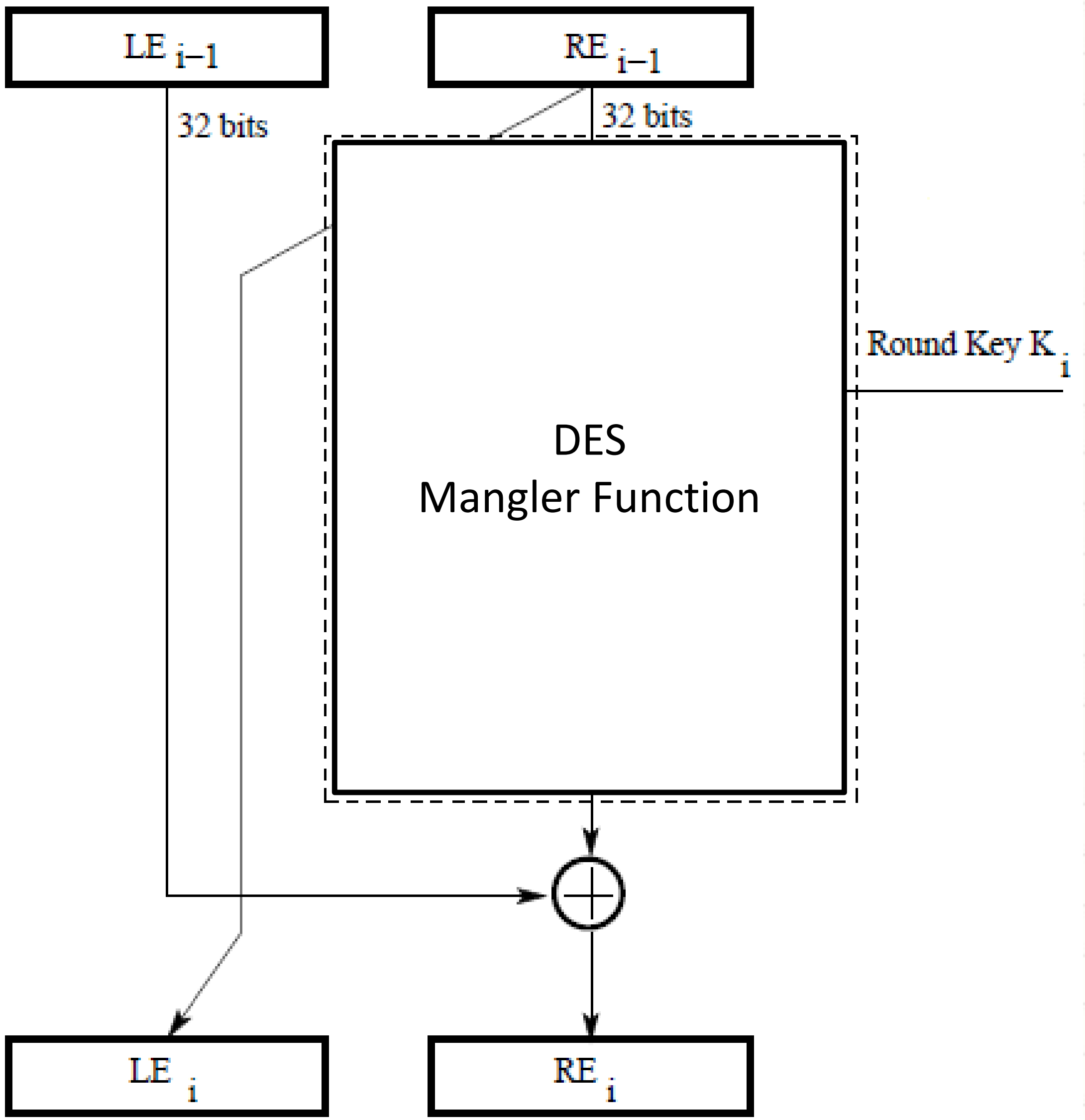
Data Encryption Standard

- Uses 56-bit key, 64-bit input block, 64-bit output block
- The key itself is specified with 8 bytes, but one bit of each byte is used as a parity check
- Uses the Feistel structure with 16 rounds
- Specific to DES is;
  - Implementation of the F function (Mangler function) in the algorithm
  - How round keys are derived from the main encryption key

- Round keys are generated from the main key by a sequence of permutations
- Each round key is 48 bits in length

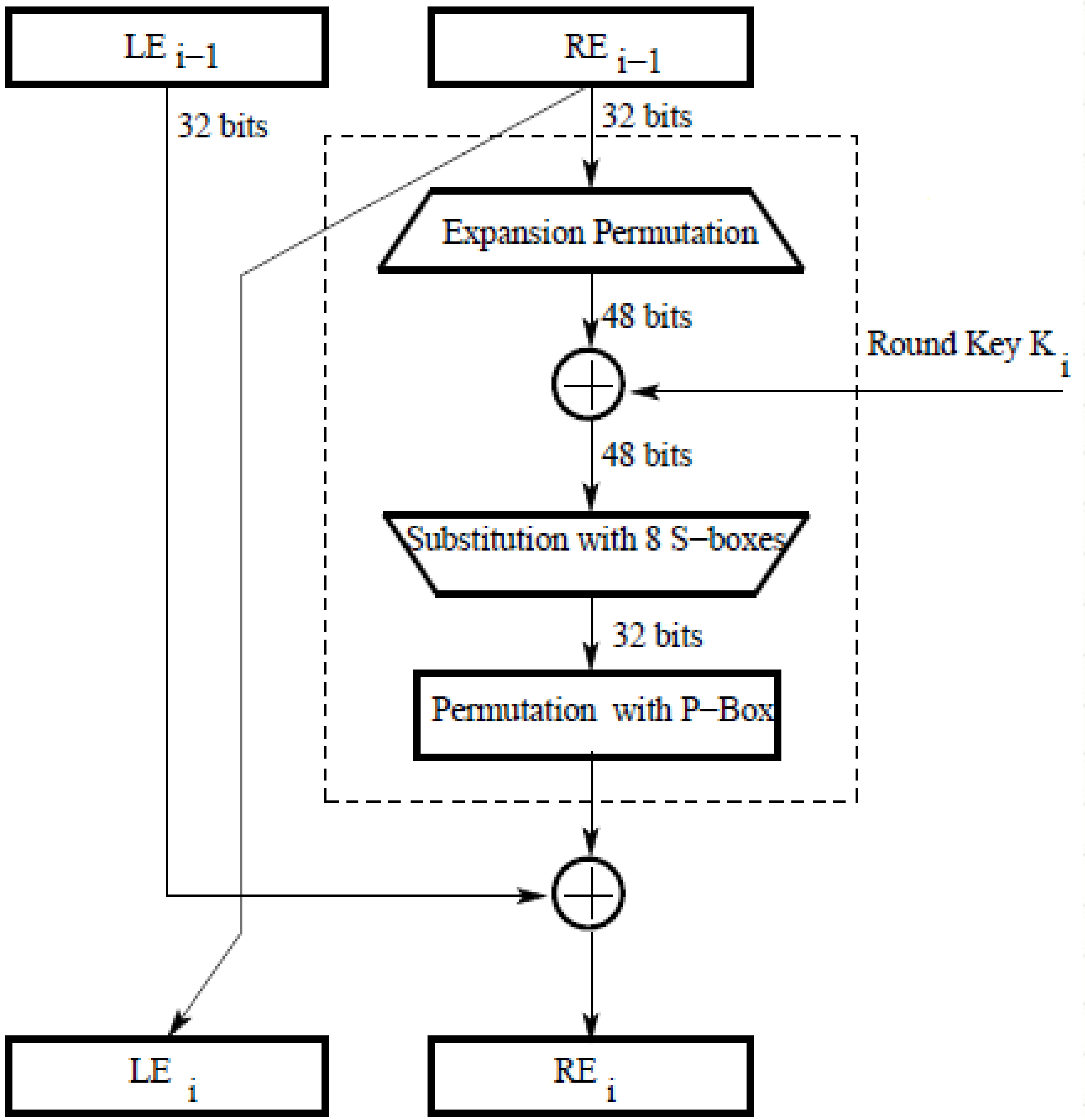
Data Encryption Standard

- Single round of DES



Data Encryption Standard

- Single round of DES



Data Encryption Standard

The F function

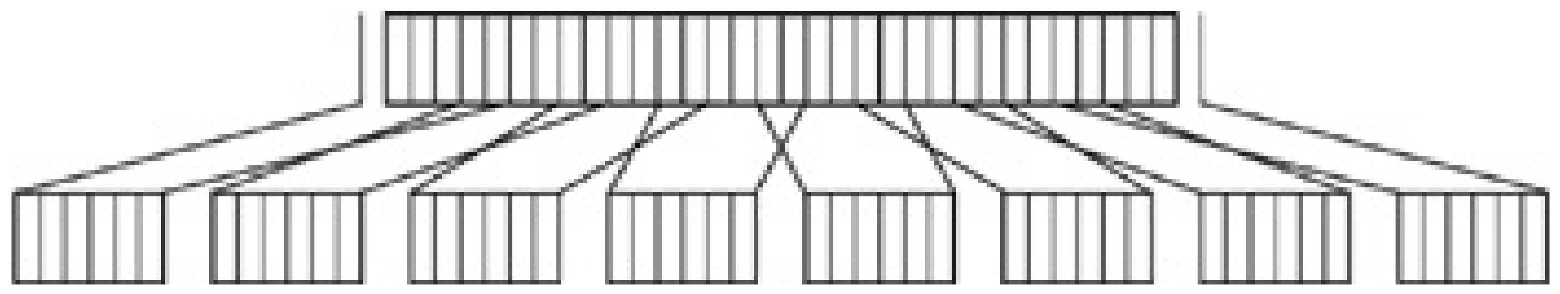
- 32-bit right half of the 64-bit input data block is expanded by into a 48-bit block. This is referred to as the expansion permutation step (**E-step)**
- 56-bit key is permuted/contracted to yield a 48-bit round key
- 48 bits of the expanded output produced by the E-step are XORed with the round key. This is referred to as **key mixing**
- Output (of the above step) is broken into eight 6-bit words. Each 6-bit word goes through a **substitution step**, where it is replaced with a 4-bit word. The substitution is carried out with an S-box
- 32-bit word (resulted from the above step) then go through a **P-box based permutation**

Data Encryption Standard

- Output from the F function is then XORed with the left half (32-bit) of the 64-bit input block
- Output of this XORing operation gives the right half block for the next round
- Left half of the next round is the right half of the previous round

Data Encryption Standard

Expansion permutation step (E-step)

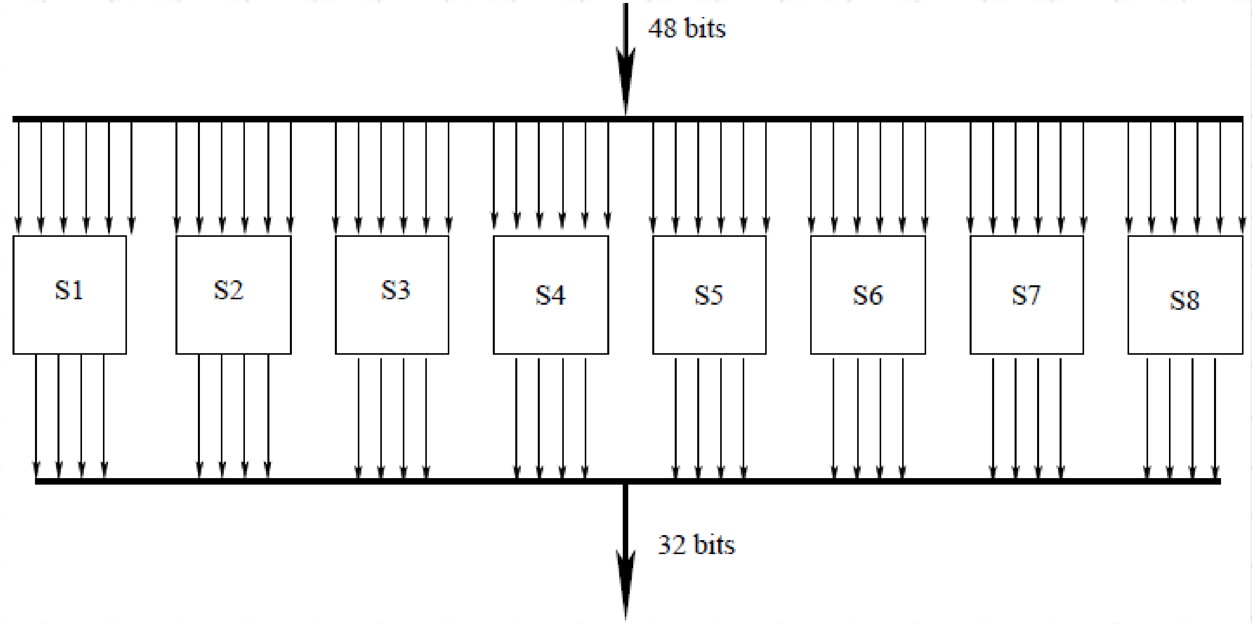


- ❑ Divide the 32-bit block into eight 4-bit words
- ❑ Attach an additional bit on the left to each 4-bit word that is the last bit of the previous 4-bit word
- ❑ Attach an additional bit to the right of each 4-bit word that is the beginning bit of the next 4-bit word

Data Encryption Standard

S-box substitution step

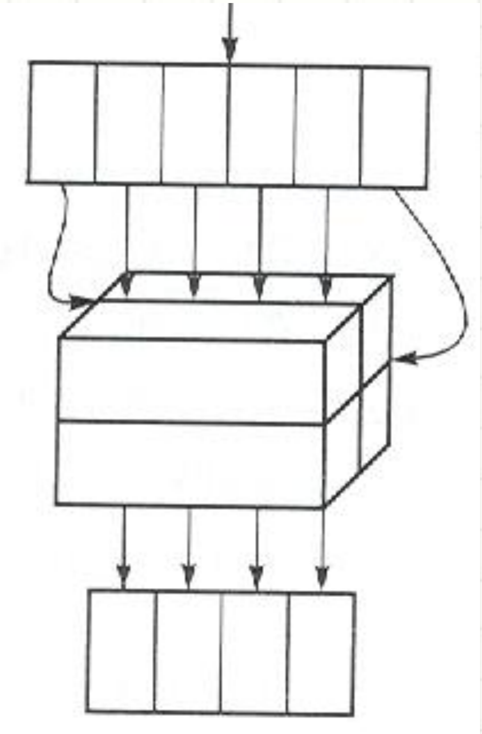
- 48-bit input word is divided into eight 6-bit words
- Each 6-bit word fed into a separate S-box
- Each S-box produces a 4-bit output
- 8 S-boxes together generate a 32-bit output



Data Encryption Standard

<u>S-box substitution step</u>

- Each of the eight S-boxes consists of a 4×16 table lookup
- First and last bit of each 6-bit input word are decoded into one of 4 rows and the middle 4 bits decoded into one of 16 columns for the table lookup



	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
00																
01																
10																
11																

Data Encryption Standard

- Eight S-boxes, S1 through S8

The 4 × 16 substitution table for S-box S_1

14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

S-box S_2

15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9

S-box S_3

10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12

S-box S_4

7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15
13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9
10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4
3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14

S-box S_5

2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3

S-box S_6

12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11
10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8
9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6
4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13

S-box S_7

4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1
13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6
1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2
6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12

S-box S_8

13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7
1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2
7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8
2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11

Data Encryption Standard

P-Box Permutation step

- 32-bit (output from previous step) goes through the last step which is permutation with P-Box

P-Box Permutation							
16	7	20	21	29	12	28	17
1	15	23	26	5	18	31	10
2	8	24	14	32	27	3	9
19	13	30	6	22	11	4	25

Data Encryption Standard

P-Box Permutation step

- 32-bit (output from previous step) goes through the last step which is permutation with P-Box

P-Box Permutation							
16	7	20	21	29	12	28	17
1	15	23	26	5	18	31	10
2	8	24	14	32	27	3	9
19	13	30	6	22	11	4	25

16	1
7	2
20	3
21	4
29	5
12	6
28	7
17	8
1	9
15	10
23	11
26	12
5	13
18	14
31	15
10	16

2	17
8	18
24	19
14	20
32	21
27	22
3	23
9	24
19	25
13	26
30	27
6	28
22	29
11	30
4	31
25	32

Data Encryption Standard

The DES Key: Generating the Round Keys

- Initial 56-bit key may be represented as 8 bytes, with the last bit (the least significant bit) of each byte used as a parity bit
- Relevant 56 bits are subject to a permutation at the beginning before any round keys are generated, referred to as Permutation Choice 1

Permutation Choice 1						
57	49	41	33	25	17	9
1	58	50	42	34	26	18
10	2	59	51	43	35	27
19	11	3	60	52	44	36
63	55	47	39	31	23	15
7	62	54	46	38	30	22
14	6	61	53	45	37	29
21	13	5	28	20	12	4

Data Encryption Standard

The DES Key: Generating the Round Keys

- At the beginning of each round
  - Divide the 56 relevant key bits into two 28-bit halves and circularly shift to the left each half by one or two bits, depending on the round

- In rounds 1, 2, 9, and 16 rotate 1-bit left
- In other rounds rotate 2-bit left

Data Encryption Standard

The DES Key: Generating the Round Keys

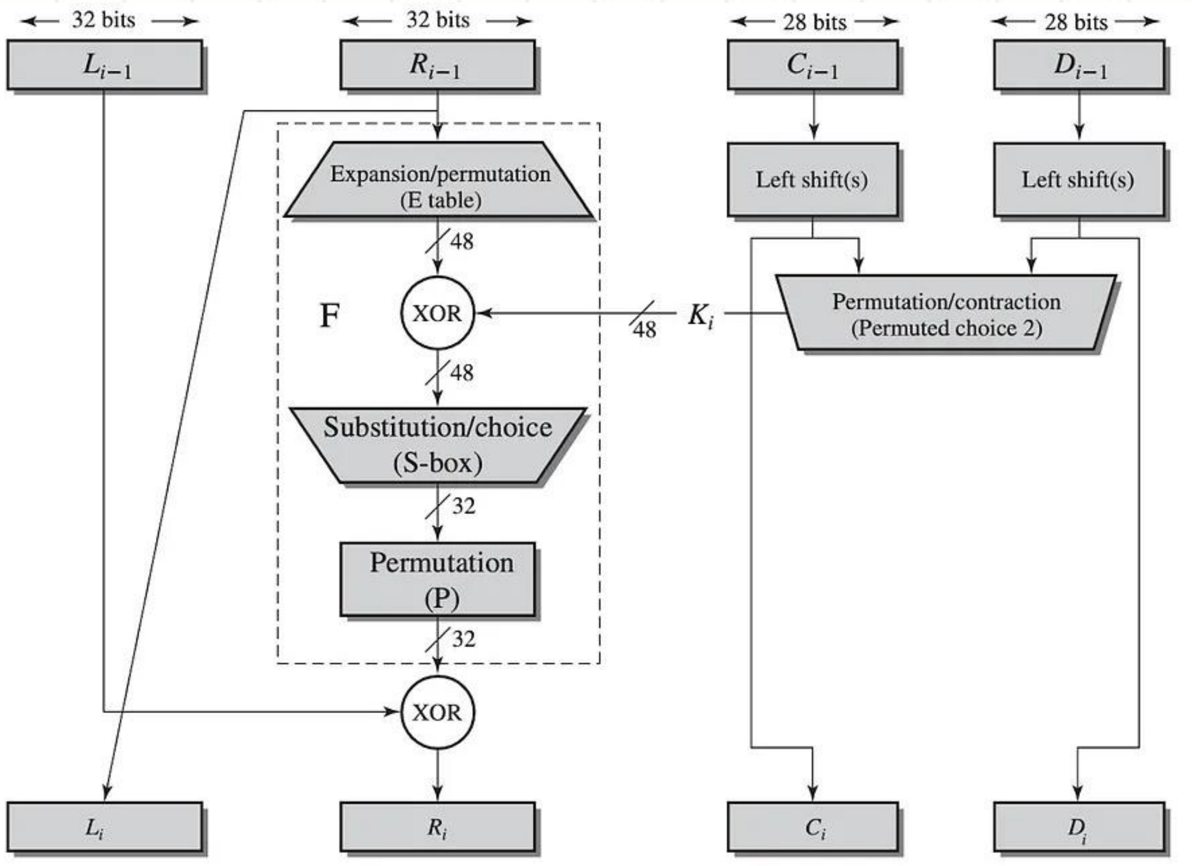
- Join together the two halves and apply a 56 bit to 48 bit contracting permutation, referred to as Permutation Choice 2
- Resulting 48 bits constitute round key

Permutation Choice 2							
14	17	11	24	1	5	3	28
15	6	21	10	23	19	12	4
26	8	16	7	27	20	13	2
41	52	31	37	47	55	30	40
51	45	33	48	44	49	39	56
34	53	46	42	50	36	29	32

- Two halves of the encryption key generated in each round are fed as the two halves going into the next round

Data Encryption Standard

Single Round of DES



Data Encryption Standard

What makes DES a strong cipher

- Substitution step is very effective
- It has been shown that if you change just one bit of the 64-bit input data block, on the average that alters 34 bits of the ciphertext block

- Round keys generation from the encryption key is also very effective
- It has been shown that if you change just one bit of the encryption key, on the average that changes 35 bits of the ciphertext

- Both effects are referred to as the avalanche effect

Data Encryption Standard

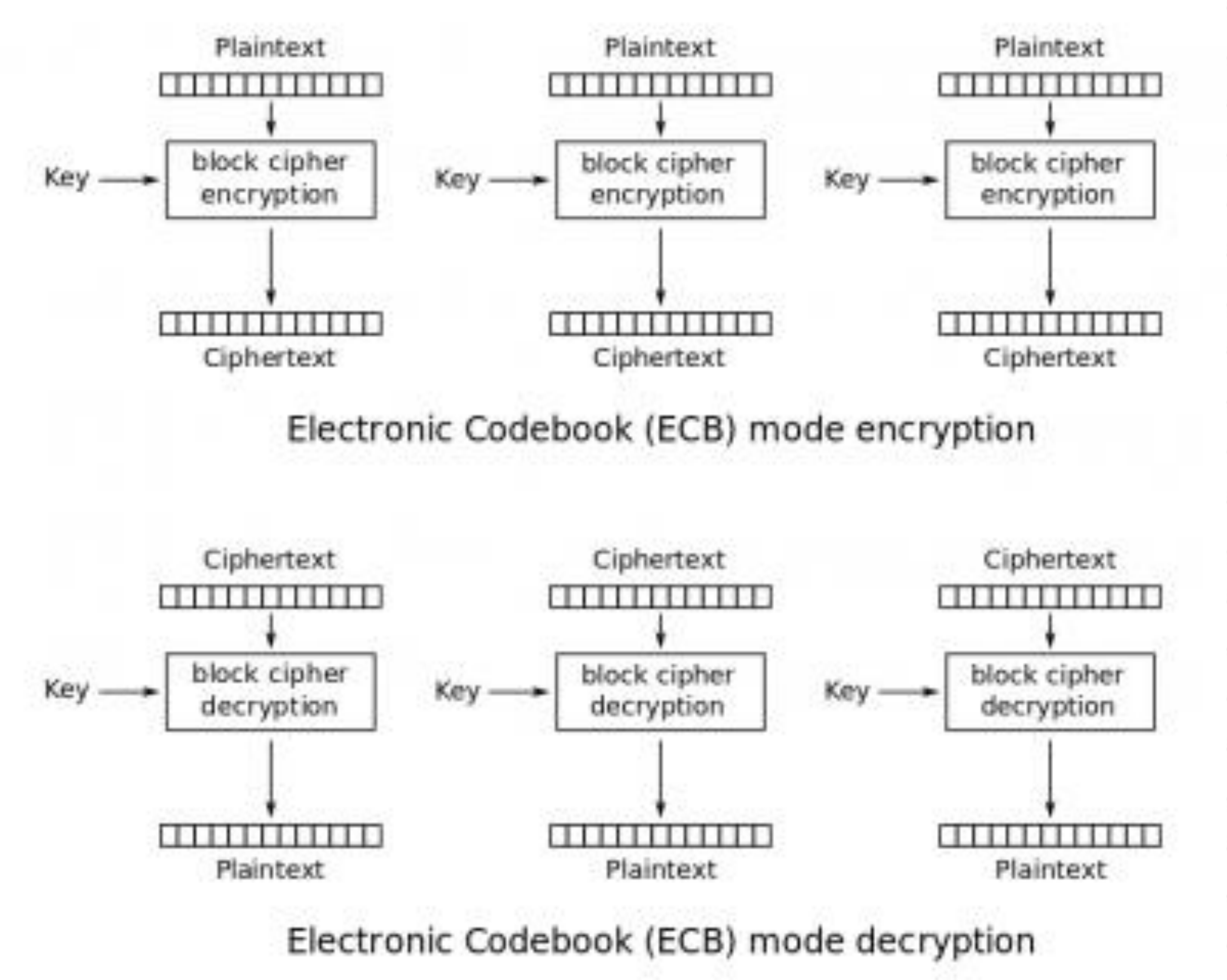
What makes DES a strong cipher

- 56-bit encryption key means a key space of size $2^{56} \approx 7.2 \times 10^{16}$
- On the average, you'd need to try half the keys in a brute-force attack
- A machine able to process 1000 keys per microsecond would need roughly 13 months to break the code
- However, a parallel-processing machine trying 1 million keys simultaneously would need only about 10 hours

Data Encryption Standard

DES Modes

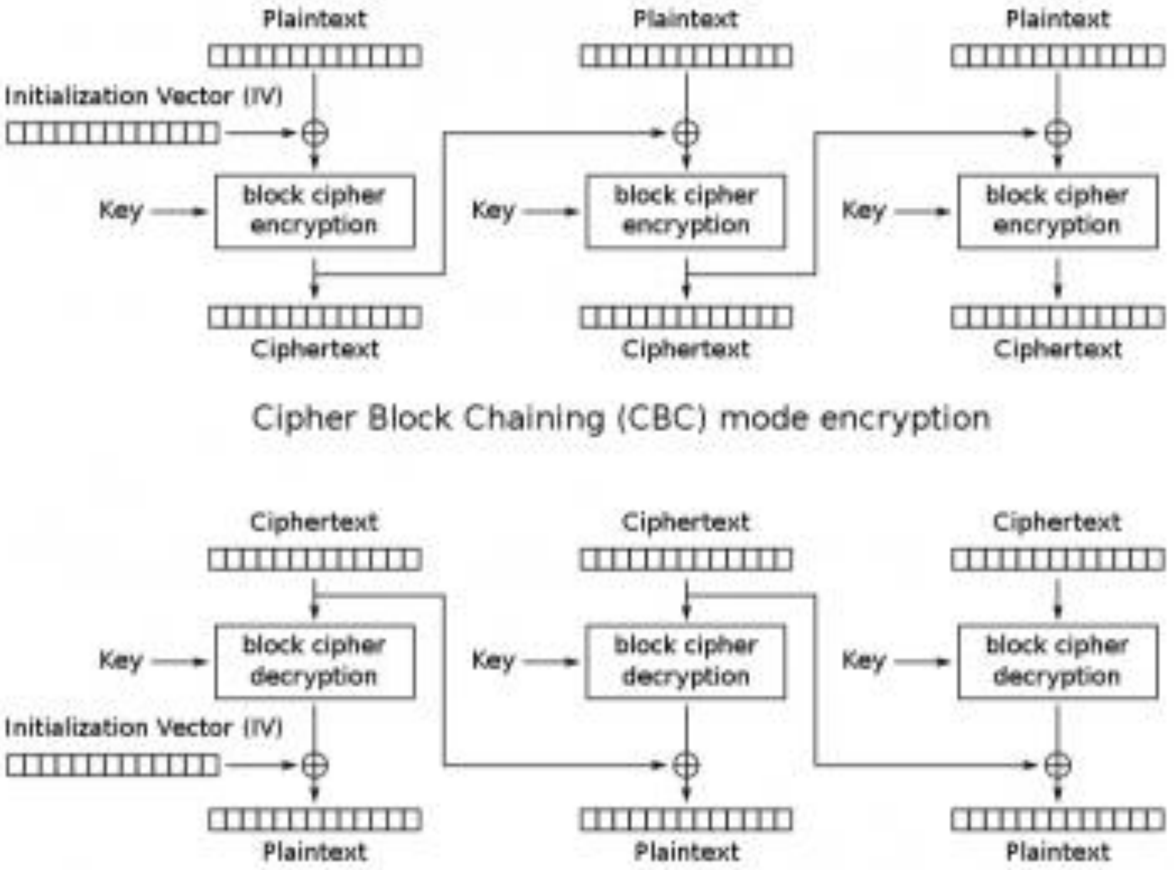
- ECB



Data Encryption Standard

DES Modes

- CBC



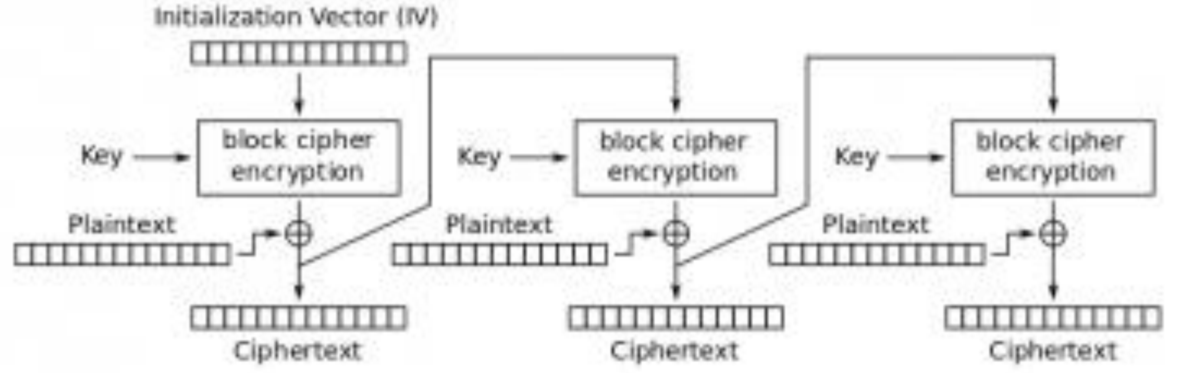
Cipher Block Chaining (CBC) mode encryption

Cipher Block Chaining (CBC) mode decryption

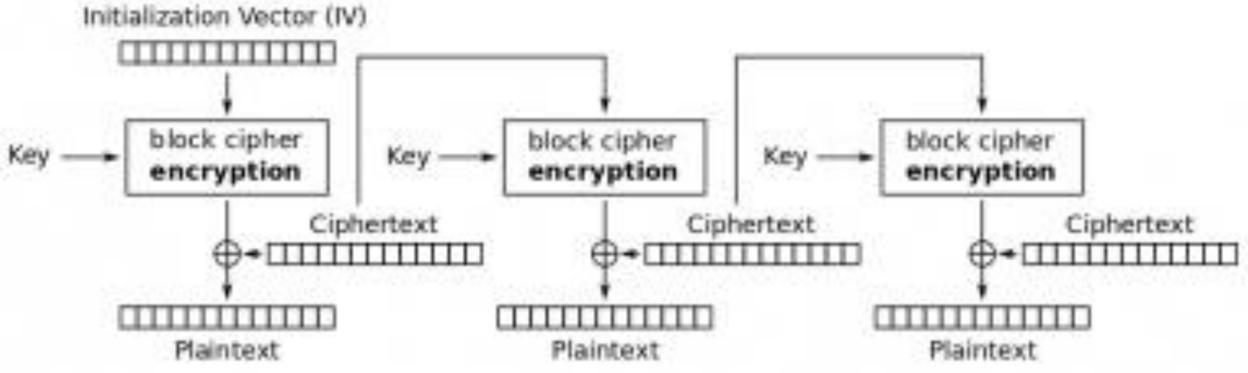
Data Encryption Standard

DES Modes

- CFB



Cipher Feedback (CFB) mode encryption

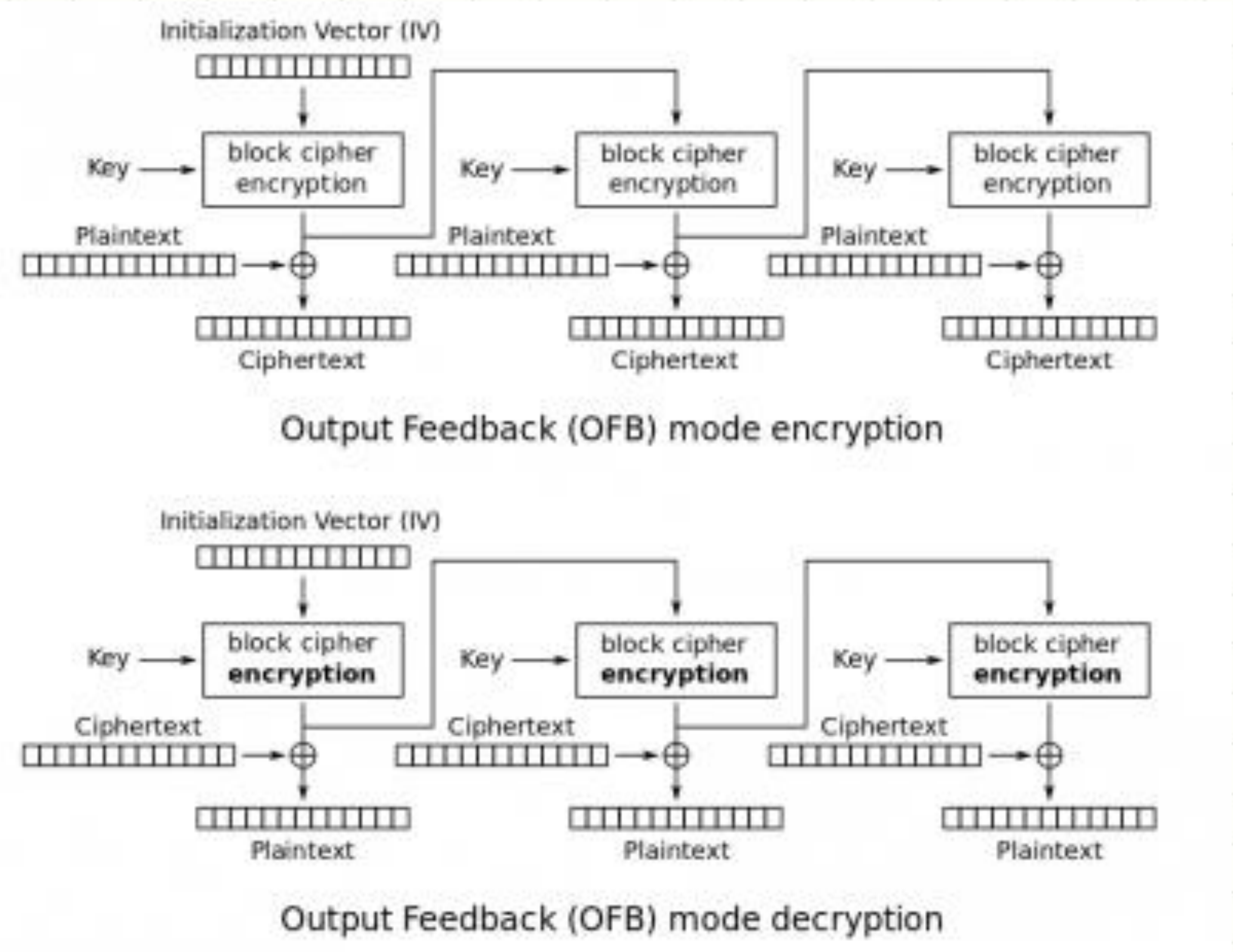


Cipher Feedback (CFB) mode decryption

Data Encryption Standard

DES Modes

- OFB



Data Encryption Standard

- DES was broken in 1998 by Electronics Frontiers Foundation (EFF)
- It took 56 hours on a specially designed machine (Deep Crack, $220,000), trying 90 billion keys a second, to break the code
- This resulted in NIST issuing a new directive that year that required organizations to use **Triple DES (3DES)**, that is, three consecutive applications of DES
- 3DES was found to be not as strong as originally believed
- So, NIST initiated the development of new standards for data encryption and the result is **Advanced Encryption Standard (AES)**

Data Encryption Standard

- Triple DES continues to enjoy wide usage in commercial applications even today

- DES was broken because its short key size permits rapid key-search
- But DES is a very strong design as evidenced by the fact that there are no practical attacks that exploit its structure

Data Encryption Standard

- DES Cryptanalysis

- DES has 4 weak and 12 semi weak keys
- Brute force attack on the keys
- Differential cryptanalysis - Biham-Shamir
- Linear cryptanalysis - Mitsuru Matsui

Data Encryption Standard

- How Block Ciphers could be improved
  - Variable key length
  - Mixed operators: use more than one and/or, this can provide non-linearity
  - Key-dependent S-boxes
  - Lengthy key schedule algorithm
  - Variable plaintext/ciphertext block length
  - Variable number of rounds
  - Operation on both data halves each round
  - Variable f function (varies from round to round)

International Data Encryption Algorithm

- IDEA is a block cipher designed by Xuejia Lai and James Massey of ETH Zurich in 1991
- It is patented in a number of countries but can be used by anyone for non-commercial use
- Operates on 64-bit blocks, using a 128-bit key
- 8 identical transformations (rounds) and 1 output transformation (the half-round), a total of 8.5 rounds
- Each of the 8 round use 6 sub-keys, while the half-round uses 4
- 52 total sub-keys of 16-bit in length each

International Data Encryption Algorithm

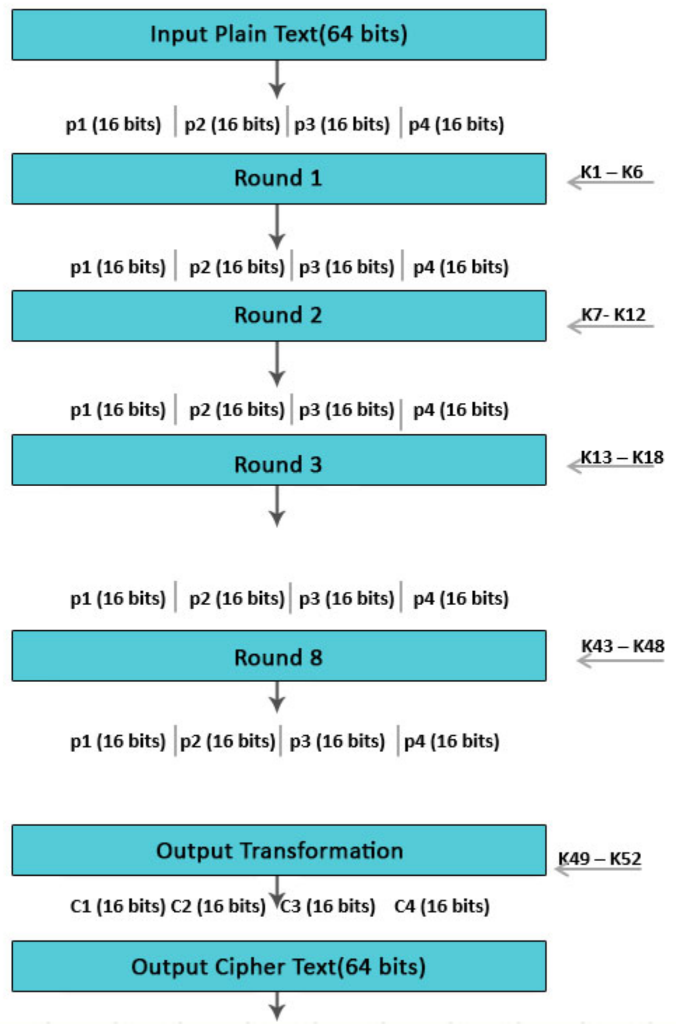
- 64-bit input plaintext block is divided into 4 parts (16 bits each) [p1 to p4]
- Key = 128 bits, in each round, 6 sub-keys will be produced (total 8 rounds)
- Each sub-key = 16 bits, sub-keys are used on the 4 input blocks p1 to p4
- Last round use 4 sub-keys
- Output is 4 blocks of 16 bits ciphertext [c1 to c4], concatenated to create the 64-bit ciphertext block

International Data Encryption Algorithm

- 128-bit key is used to generate 52 16-bit subkeys
- Initially, the first 8 subkeys are directly taken from the first 128 bits of the key
- For subsequent subkeys, the key is rotated 25 bits to the left before extracting the next 16 bits

International Data Encryption Algorithm

- 8.5 rounds



International Data Encryption Algorithm

- Each round operations

Step 1: Multiply* p1 and K1

Step 2: Add* p2 and K2

Step 3: Add* p3 and K3

Step 4: Add* p4 and K4

Step 5: XOR the results step 1 and 3

Step 6: XOR the results step 2 and 4

Step 7: Multiply *the results step 5 with K5

Step 8: Add*the results step 7 and step 9

Step 9: Multiply *the results of step 8 and K6

Step 10: Add *the results of step 7 and step9

Step 11: XOR the results of step 1 and step9

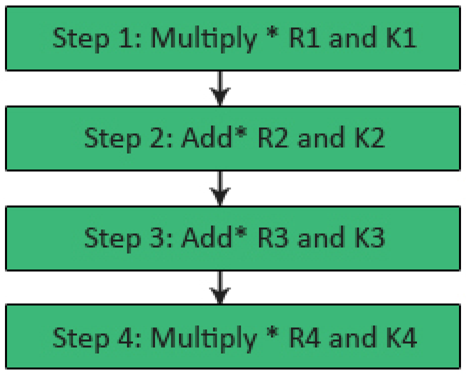
Step 12: XOR the results of step 3 and step9

Step 13: XOR the results of step 2 and step 10

Step 14: XOR the results of step 4 and step 10

International Data Encryption Algorithm

- Last round operations



References

- DES official documentation

www.itl.nist.gov/fipspubs/fip46-2.htm

- DES Cipher Algorithm

http://www.herongyang.com/Cryptography/DES-Algorithm-Data-Encryption-Standard.html

- Test DES encryption

https://www.openssl.org/

- "Cryptography and Network Security: Principles and Practice", 6th edition, by William Stallings
- "Network Security: Private Communication in a Public World", by Charlie Kaufman, Radia Perlman, Mike Speciner (chapter 3)

References

- Simplified DES

http://mercury.webster.edu/aleshunas/COSC%205130/G-SDES.pdf

- Another example of Simplified DES

http://homepage.smc.edu/morgan_david/vpn/des.htm

- DES encryption/decryption example

http://homepages.gac.edu/~holte/courses/mcs150-J01/documents/DESexample.html

THANKS